



June 12, 2023

Stephanie Weiner  
Acting Chief Counsel  
National Telecommunications and Information Administration  
U.S. Department of Commerce  
1401 Constitution Avenue NW  
Room 4725  
Washington, DC 20230

*Submitted electronically at [www.regulations.gov](https://www.regulations.gov)*

RE: AI Accountability Policy Request for Comment (Docket No. NTIA-2023-0005)

Dear Ms. Weiner:

Equifax appreciates the opportunity to submit comments regarding the National Telecommunications and Information Administration's Request for Comment on how to increase public trust in AI solutions. Equifax is a global information solutions company that uses unique data, innovative analytics, technology, and industry expertise to power organizations, government agencies, and individuals around the world by transforming knowledge into insights that help make more informed business and personal decisions. Data scientists in the global Equifax Data & Analytics Lab work to better connect the company's unique data with the specific needs of customers through a relentless pursuit of new innovation in predictive analytics, including through the use of artificial intelligence (AI) and machine learning (ML) to develop innovative credit scoring systems. Equifax maintains nineteen patents granted in the United States, and more awarded worldwide, in the area of AI and ML, with particular respect to credit scoring systems, demonstrating an industry leading expertise in the subject matter.

The National Telecommunications and Information Administration (NTIA) seeks input on several questions regarding how to increase public trust in AI systems, including what kind of data access is necessary to conduct audits and assessments, and what different approaches might be needed in different industry sectors. This comment focuses on published research by economists to document that the use of inappropriate data in fair lending and credit decisioning analyses can lead to contradictory results. We accomplish this by juxtaposing the results of several published studies using self-reported demographic data against results from a series of papers using aggregated or proxy measures attempting to control and understand differences (if any) for certain demographic groups.

### **Background**

Different implementations of AI make use of different types and levels of data. Marketing applications, for example, seek to discover populations of individuals that are likely to purchase a retailer's products. These applications want to identify large groups of people who are invited to apply through marketing

stimuli. The target audience in these cases is a broad swath of the population, and marketers are looking for large groups of people who tend to respond similarly to marketing messages.

Other applications, including credit scoring, are much more narrow and highly regulated. Equifax develops machine learning credit scoring systems based on individual behaviors reported by data furnishers. The data used resides on the consumers' credit report. These behaviors measure consumer responses to the consumer's contractual credit obligations. It would be inappropriate to evaluate a consumer's credit worthiness based on the consumer's neighbors' behavior. In general, the data used to assess a model's appropriateness for purpose should only use data collected at the same level of analysis used by the AI system.

### **Individual Level Analysis Must Use Individual-Level Data**

Economists at the Federal Reserve have demonstrated what happens when appropriate or inappropriate data are used to attempt to identify disparities in credit scoring and lending. Multiple lines of research results conducted over many years show that when analysis is conducted at the individual level, that assessments must also make use of individual-level protected class data in order accurately to assess disparities.

For instance, in August 2007, the Federal Reserve Board of Governors presented the *Report to the Congress on Credit Scoring and Its Effects on the Availability and Affordability of Credit* (hereinafter, *The Report*) as required by Congress.<sup>1</sup> *The Report* uses self-reported demographic data merged from Social Security applications, resulting in a very high match rate to the credit population under consideration, and highly reliable individual-level data on protected classes of interest. These results have been published two additional times, in 2010 and in 2012, and have not empirically been refuted.<sup>2</sup> In 2018, Geng Li reported a similar lack of disparity in gender-related differences in credit use and credit scores<sup>3</sup> In both cases, the data used to assess disparities was self-reported demographic information. In the cases of Avery, Brevoort, and Canner, Social Security application data were used, and in Li (2018), self-reported survey data was used.

### **Aggregate and Proxy Data for Individual Level Analysis has Proved Problematic**

In all cases where disparities are meant to be assessed, knowledge of a consumer's membership in a protected class is required. In the absence of self-reported individual data, researchers have turned to imputation of class membership using various statistical measures. Such measures include assignment of individual race using proportions of Black residents in an individuals' census tract (Laufer and Paciorek, 2016), using block-level census data to infer individual membership for race, education, and

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<sup>1</sup> Board of Governors of the Federal Reserve, 2007. *Report to the Congress on Credit Scoring and Its Effects on the Availability and Affordability of Credit*. [www.federalreserve.gov/boarddocs/rptcongress/creditscore/creditscore.pdf](http://www.federalreserve.gov/boarddocs/rptcongress/creditscore/creditscore.pdf)

<sup>2</sup> See: Avery, Robert B., Kenneth P. Brevoort, and Glenn B. Canner, 2010 "Does Credit Scoring Produce a Disparate Impact?" Finance and Economics Discussion Series (FEDS), 2010-058.

[www.federalreserve.gov/pubs/feds/2010/201058/201058pap.pdf](http://www.federalreserve.gov/pubs/feds/2010/201058/201058pap.pdf), and Avery, Brevoort, and Canner, 2012, "Does Credit Scoring Produce a Disparate Impact?", *Real Estate Economics*, American Real Estate and Urban Economics Association, v40, pp 65-114, December. The latter paper appeared in a peer-reviewed academic journal.

<sup>3</sup> Li, Geng, 2018. "Gender-Related Differences in Credit Use and Credit Scores" Finance and Economics Discussion Series Notes, p8.

<https://www.federalreserve.gov/econres/notes/feds-notes/gender-related-differences-in-credit-use-and-credit-scores-20180622.html>

income (Dokko, Li, and Hayes, 2015), collapsing ethnicity responses into fewer categories because of small sample sizes (Edelberg, 2007), and other purely statistical proxy measures for ethnicity including Bayesian Improved Surname Geocoding (BISG) introduced to the financial industry by the Consumer Financial Protection Bureau in 2014.<sup>4</sup>

In some cases using proxy or aggregated protected class data disparities are reported, but may be explained by omitted variables or model misspecification. In other research using proxy data, no disparities are reported (Bhutta and Hizmo, 2020).<sup>5</sup> In general, there is substantial doubt regarding the veracity of reported results purporting to measure disparities in credit markets using aggregated or proxy data.

When using aggregate data to identify protected class status, one is either making a probabilistic assignment, or using that probability measure to make a definitive assignment. Both methods introduce problems in analyses. Probabilistic assignment occurs when the researcher assigns to an individual a measure based on the proportion of a specific class in some geography. For example, Dokko, Li, and Hayes (2015) use block-group level averages to

“approximate individual level demographic and socioeconomic characteristics, such as race, education, and income. For example, we use the *share of adults in a block group who are white ...to approximate individual race ...*, and the *census block group median income to approximate individual income.*” (pp. 10-11, emphasis added).

The measures being assigned to individuals in the case above are aggregated. For income, every individual in a census block group receives an identical value - the median, which is inappropriate for almost every individual in the block group. For race, every individual in the census block group receives a percentage “White”. Such a measure is incorrect for every individual in the analysis, unless one lives in a block comprised of one race. The use of such data can lead to the ecological fallacy, which occurs when inferences about individuals are based on inferences about the group to which those individuals belong.<sup>6</sup> The fallacy can lead to incoherent results in analyses where the actual result is the opposite of what is expected. Simpson’s Paradox can occur when subgroup information is aggregated and used in individual-level analyses. Simpson’s Paradox may explain one result reported by Larrimore, Shuetz, and

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<sup>4</sup> Laufer, Steven and Andrew Paciorek, 2016. “The Effects of Mortgage Credit Availability: Evidence from Minimum Credit Score Lending Rule”. Finance and Economics Discussion Series, 2016-098. <https://www.federalreserve.gov/econres/feds/the-effects-of-mortgage-credit-availability-evidence-from-minimum-credit-score-lending-rules.htm>, Jane Dokko, Geng Li, and Jessica Hayes, “Credit Scores and Committed Relationships”. Finance and Economics Discussion Series, 2015-081. [www.federalreserve.gov/econres/feds/credit-scores-and-committed-relationships.htm](http://www.federalreserve.gov/econres/feds/credit-scores-and-committed-relationships.htm), Wendy Edelberg, 2007, “Racial Dispersion in Consumer Credit Interest Rates”, Finance and Economics Discussion Series, 2007-28. [www.federalreserve.gov/econres/feds/racial-dispersion-in-consumer-credit-interest-rates.htm](http://www.federalreserve.gov/econres/feds/racial-dispersion-in-consumer-credit-interest-rates.htm), and Consumer Financial Protection Bureau, 2014 “Using publicly available information to proxy for unidentified race and ethnicity”, [files.consumerfinance.gov/f/201409\\_cfpb\\_report\\_proxy-methodology.pdf](https://files.consumerfinance.gov/f/201409_cfpb_report_proxy-methodology.pdf).

<sup>5</sup> Bhutta, Neil and Aurel Hizmo, 2020. “Do Minorities Pay More for Mortgages?”, Federal Reserve Board: Finance and Economics Discussion Series, 2020-007. <https://www.federalreserve.gov/econres/feds/do-minorities-pay-more-for-mortgages.htm>. Notably, this paper uses HMDA from 2014-2015.

<sup>6</sup> Robinson, WS, 1950. “Ecological Correlations and the Behavior of Individuals”, *American Sociological Review*, v15, No 3, pp. 351-357, June 1950, reprinted in *International Journal of Epidemiology*, 2009, v38, pp 337-341.

Dodini (2016) where the authors use aggregated information assigned to individuals to investigate barriers to homeownership among young people:<sup>7</sup>

“More puzzling, however, is that county-level measures of housing affordability generally are not significantly predictive of the perceived barriers to homeownership among young renters, and the one instance where they are – considering the impact of the price-rent ratio on renting because it is cheaper – the direction of the effect is the opposite of that expected.” (p18)

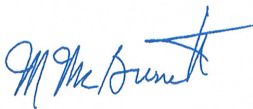
All data used for analysis should be collected at the appropriate level defined by the analytical goal - individual level versus aggregated level.

If the use of aggregated demographics to investigate disparities is problematic, perhaps the measures can be improved. This is the basis for use of Bayesian Improved Surname Geocoding (BISG), introduced by the Consumer Financial Protection Bureau in 2014 and subsequently used to evaluate disparities in certain auto lending decisions. BISG, as introduced, uses information about individuals' last names and their frequency of occurrence in census tracts to infer ethnicity. That inference was then used in analyses of certain lending outcomes. BISG assigns a definitive value for ethnicity to every consumer as long as sufficient data exists to cross a threshold value. The usual threshold is 0.80. This leads to significant misidentification of individual ethnicities, as has been empirically demonstrated twice.<sup>8</sup> The impact of wrongly identifying individuals' ethnicity introduces measurement error into analyses, which additionally can have a significant deleterious impact on empirical analysis.

The NTIA introduced this request for comment to help identify gaps and barriers to creating adequate accountability for AI systems, including fairness in outcomes. As the examples in this comment demonstrate, for an AI model to be trustworthy and accountable it must not rely on aggregate or proxy data. The NTIA and other government agencies examining the use of AI and ML should take this into consideration when developing standards for the audit or assessment of AI and ML models used in the credit ecosystem.

Thank you for the opportunity to offer these insights. Please contact me with any questions at michael.mcburnett1@equifax.com.

Sincerely,



Michael McBurnett  
Distinguished Scientist  
Equifax Data and Analytics



Nick Oldham  
Chief Privacy and Compliance Officer

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<sup>7</sup> Larrimore, Jeff, Jenny Shuetz, and Samuel Dodini, 2016. “What are the Perceived Barriers to Homeownership for Young Adults?”, Federal Reserve Board: Finance and Economics Discussion Series, 2016-021, p.18.

<sup>8</sup> Baines, Arthur P. and Marsha J. Courchane, 2014. “Fair Lending: Implications for the Indirect Auto Finance Market” Charles River Associates (see Appendix G), and Woodford, Terry and Michael McBurnett, 2021 “Assessing Disparate Impact for Machine Learning Credit Scoring Models: The Available Data Do Not Meet Analytical Requirements”, presented at the 2021 Credit Scoring and Credit Control Conference, Edinburgh, Scotland, August 2021.